Dust and Pollen Events Show Similar Aerosol Optical Properties but Distinct Particle Size-Distribution Characteristics in a Finnish Boreal Forest

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Abstract

Dust and pollen events both perturb boreal aerosol populations, but their in-situ optical signatures can overlap. Here, we analyse dust and pollen events at the Station for Measuring Ecosystem–Atmosphere Relations II (SMEAR II), Hyytiälä, Finland, using size-resolved aerosol optical properties, particulate matter (PM) mass, and aerosol particle size distributions (APSDs). We separate the overall event effect, event detection, and event typing. The overall event effect describes how dust and pollen modify aerosol optical properties, particle size distributions, and PM mass. Event detection describes separation between dust/pollen event conditions and non-event background conditions. Event typing describes separation between dust events and pollen events within the event subset. Both event types enhanced aerosol scattering relative to background conditions, but scattering coefficients and scattering Ångström exponents (SAEs) did not robustly distinguish dust from pollen. Event detection strengthened with particle size and was strongest for super-PM_{10} mass, defined here as the derived mass fraction above 10 μm, whereas the clearest mass-based dust–pollen typing signal occurred for $PM_{1\text{–}10}$ mass in the present event set. Among the APSD-derived metrics, accumulation-mode cross-sectional area, S_{acc}, showed the clearest temporally controlled dust–pollen contrast, with higher anomalies during dust events than pollen events. However, event-declustered uncertainty indicates that this result should be interpreted cautiously because of the small number of independent pollen events. These results show that dust–pollen separation in boreal aerosol observations is controlled less by bulk optical enhancement than by size-resolved redistribution of aerosol cross-sectional area and mass.

Introduction

Mineral dust and pollen episodically perturb atmospheric aerosol populations by adding particles, modifying particle-size distributions, and changing aerosol optical and mass properties. In boreal environments, these perturbations occur on top of a background aerosol population shaped by frequent new-particle formation (NPF), seasonal biogenic emissions, and long-term variability in particle number, chemical composition, and optical properties.[1,4,7,10,13] At the Station for Measuring Ecosystem–Atmosphere Relations II (SMEAR II), Hyytiälä, Finland, this background provides a well-characterized setting for testing how episodic dust and pollen events modify in-situ aerosol properties.

Dust and pollen differ in source process, transport history, composition, morphology, and size structure, but they can still produce overlapping aerosol responses at the receptor site. Both can increase coarse and supermicron aerosol abundance and thereby enhance scattering and particulate matter (PM) mass. Atmospheric pollen can be detected optically by lidar depolarization, and long-range transported dust has been documented to reach Finland from Saharan, Aral-Caspian, and Middle Eastern source regions.[2,3,14,20] However, in-situ optical variables are not uniquely source-specific because scattering depends on aerosol loading, particle-size distribution, refractive index, hygroscopic growth, and mixing state.[16,17,19]

This creates a central interpretation problem for boreal aerosol event studies. A variable may respond strongly during dust or pollen events and therefore indicate that an aerosol perturbation has occurred, but the same variable may still be unable to identify which event type occurred. In other words, the effect of an event, the detection of an event, and the typing of an event are related but distinct questions. The overall event effect asks how dust and pollen modify the aerosol population. Event detection asks whether dust- or pollen-affected observations differ from non-event background conditions. Event typing asks whether dust-event observations can be distinguished from pollen-event observations within the event subset.

The present study was therefore guided by three research questions:

1. What is the effect of dust and pollen events on aerosol optical properties, particle size distributions, and PM mass concentrations at SMEAR II?

2. Which variables derived from in-situ aerosol observations show a statistically distinguishable and physically interpretable difference between non-event background conditions and dust/pollen event conditions?

3. Which of these variables also show a statistically distinguishable and physically interpretable difference between dust events and pollen events?

These questions structure the analysis. To answer the first question, we examine how dust and pollen events modify aerosol optical properties, size-resolved PM mass, and aerosol particle size distributions (APSDs) relative to the boreal background. To answer the second question, we quantify event detection by comparing the pooled dust/pollen event class with non-event background observations using robust median shifts, confidence intervals, and rank-based effect sizes. To answer the third question, we test whether the variables that respond to events also contain source-type information by comparing dust events directly with pollen events. This distinction is important because a variable can be useful for detecting an aerosol perturbation without being useful for distinguishing dust from pollen.

Here, we address these questions using size-resolved aerosol optical properties, PM mass, and APSD-derived number, cross-sectional-area, and volume metrics. We first evaluate whether optical, mass, and APSD-derived variables detect event occurrence relative to the non-event background. We then test whether the same variables can distinguish dust events from pollen events. Finally, we use the particle-size-distribution information to interpret why bulk optical signatures can overlap even when size-resolved mass and accumulation-mode APSD metrics show stronger event-type information.

Materials and Methods

Aerosol optical, PM-mass, and particle size-distribution measurements from SMEAR II, Hyytiälä, southern Finland, were analysed under background, pollen-event, and dust-event conditions.[1,9,13] The optical variables were derived from in-situ scattering and absorption measurements, while aerosol particle size distributions were derived from combined mobility and aerodynamic particle size-distribution measurements following long-term atmospheric aerosol measurement practice.[1,13,22] Detailed information on the measurement site, instruments, derived variables, operational size ranges, and valid-count differences is provided in the Supporting Information (Sect. S3).

Dust events were identified using the long-range transported dust-event framework of Varga et al.,[20] and pollen events were identified from SMEAR II impactor-based pollen observations. In the merged optical–PM-mass dataset, pollen events were assigned when any of the four processed impactor flag columns equalled the pollen-event code 555, and dust events were assigned using the `dust_event` indicator. Background observations were defined as non-event observations, i.e., observations classified as neither pollen nor dust. No dust–pollen overlap occurred in the merged optical–PM-mass dataset, so no class-priority rule affected the final event labels. Further details of the event-classification logic and final exclusive event counts are provided in the Supporting Information (Sect. S1 and Table S1).

The merged optical–PM-mass dataset contained 1581 sampling intervals before variable-specific missing-value filtering. The final event logic identified 1545 background observations, 21 pollen-event observations, and 15 dust-event observations, corresponding to 36 total event observations (Table S1). Dust-event observations occurred from February to August, whereas pollen-event observations occurred from April to June, with the largest number of pollen observations in May (Table S2). These counts refer to classified sampling intervals, not necessarily to statistically independent event episodes; therefore, the main text emphasizes effect sizes, confidence intervals, and consistency of direction rather than relying only on hypothesis-test significance. Variable-specific valid counts differed after filtering. For example, the main PM-mass variables had 1563 valid observations, while PM_1 $\sigma_{\mathrm{sca},550}$ and $PM_{1\text{–}10}$ $\sigma_{\mathrm{sca},550}$ had 1481 and 1474 valid observations, respectively (Table S4). The larger optical-only data availability was treated separately from the merged optical–PM-mass event statistics; the optical-only counts are reported only as instrumental availability diagnostics and are not used as the sample sizes for the merged optical–PM-mass event statistics (Sect. S2 and Table S3).

Event detection compared the pooled event class, $E = D \cup P$, with background, B, where D denotes dust and P denotes pollen. Dust–pollen typing compared D with P within the event subset. For a candidate variable X, event detection was quantified as a background interquartile range (IQR)-scaled median shift,

$$\Delta^*_{\mathrm{det}}(X) = \frac{\mathrm{median}(X|E) - \mathrm{median}(X|B)}{\mathrm{IQR}(X|B)}. \tag{1}$$

Dust–pollen typing was quantified as

$$\Delta^*_{\mathrm{typ}}(X) = \frac{\mathrm{median}(X|D) - \mathrm{median}(X|P)}{\mathrm{IQR}(X|B)}. \tag{2}$$

The direction of Δ^*_{typ} is therefore dust minus pollen. Positive values indicate larger values during dust events, while negative values indicate larger values during pollen events. Robustness was assessed using bootstrap confidence intervals for median differences, Cliff's δ, and Mann–Whitney tests.[5,21] Cliff's δ was used as a rank-based effect-size measure, where positive values indicate that the first group in the contrast tends to have larger values, negative values indicate the opposite, and values near zero indicate strong distributional overlap. The statistical definitions and unit-of-inference considerations are summarized in the Supporting Information (Sect. S4).

Particle number size distributions were constructed from differential mobility particle sizer (DMPS) and aerodynamic particle sizer (APS) measurements after applying instrument-specific size limits. The DMPS-based mobility size-distribution measurements follow the long-term atmospheric particle number size-distribution measurement framework used at SMEAR-type stations and related European aerosol networks.[22] The distributions were expressed as number, cross-sectional-area, and volume distributions. Cross-sectional-area and volume distributions were used to connect particle-size structure to scattering-relevant particle area and particulate material.[17] The main APSD-derived metrics were accumulation-mode cross-sectional area, S_{acc}; accumulation-mode volume, V_{acc}; coarse-mode volume, V_{coarse}; the volume-ratio metric, R_V; and Aitken-mode number concentration, N_{Ait}. Because the DMPS and APS characterize particle size using different equivalent diameters, APSD-derived cross-sectional-area and volume metrics should be interpreted as operational size-distribution metrics unless aerodynamic diameters are converted to geometric or volume-equivalent diameters using explicit density and shape assumptions (Sect. S3).

The APSD-derived metrics were analysed at their native time resolution and therefore had much larger timestamp-level sample sizes than the merged optical–PM-mass dataset (Sect. S7 and Table S9). Because native-resolution APSD measurements include repeated, temporally autocorrelated observations within event periods, the very small APSD p values are treated as supporting diagnostics rather than as the primary evidence. The main APSD inference is based on the direction and magnitude of the controlled median differences, their confidence intervals, their persistence after month and month-plus-hour background correction, and event-declustered or moving-block-bootstrap sensitivity checks where available (Tables S10 and S11).

Because the number of independent dust and pollen events was small, especially after event declustering, the results should be interpreted as evidence for physically interpretable event-type tendencies rather than as a definitive operational classification model.

Results and Discussion

Optical properties reveal event perturbation but weak event typing

Dust and pollen events both enhanced aerosol light scattering relative to non-event background conditions in the merged optical–PM-mass analysis (Figure 1). For PM_1 $\sigma_{sca,550}$, the standardized event–background median shift was $\Delta^*_{det} = 0.88$, with a raw median-difference confidence interval of 3.85–12.29 Mm^{-1} and Cliff's $\delta = 0.463$. $PM_{1–10}$ $\sigma_{sca,550}$ also increased under event conditions, although more weakly, with $\Delta^*_{det} = 0.81$, a confidence interval of 0.48–2.88 Mm^{-1}, and Cliff's $\delta = 0.342$. These effect sizes refer to the merged optical–PM-mass dataset, where the corresponding valid observation counts were 1481 and 1474, respectively (Table S4). The full event-detection effect-size results are provided in Table S5.

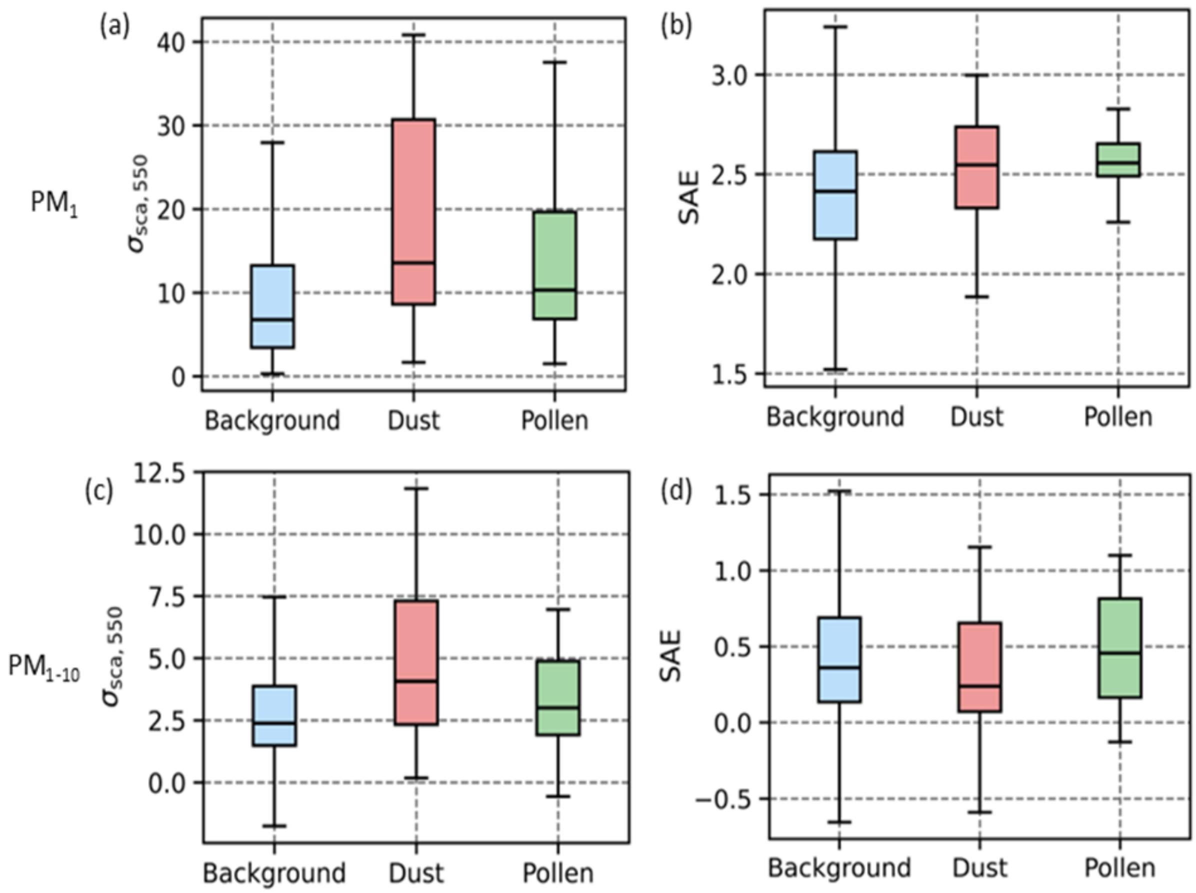

Figure 1: Aerosol optical-property response during background, dust-event, and pollen-event conditions in the merged optical–PM-mass analysis. Panels show (a) PM_1 scattering coefficient at 550 nm, $\sigma_{\mathrm{sca},550}$, (b) PM_1 scattering Ångström exponent (SAE), (c) $PM_{1\text{–}10}$ $\sigma_{\mathrm{sca},550}$, and (d) $PM_{1\text{–}10}$ SAE. Boxes show the interquartile range, horizontal lines indicate medians, and whiskers show the 5th–95th percentile range. Both dust and pollen events enhance scattering relative to background conditions, but the dust and pollen optical distributions overlap substantially, indicating that these optical variables mainly detect event occurrence rather than robustly distinguish event type.

The optical event-detection response did not translate into robust dust–pollen typing. The dust–pollen median-difference confidence intervals crossed zero for both PM_1 and $PM_{1\text{–}10}$ $\sigma_{\mathrm{sca},550}$ (Table S6). SAE also provided limited source-type information. PM_1 SAE showed a small event–background response, with $\Delta^*_{\mathrm{det}} = 0.38$ and a confidence interval of 0.08–0.21, while $PM_{1\text{–}10}$ SAE did not show a robust event-detection response (Table S5). Dust–pollen SAE typing was not robust in either size fraction (Table S6).

This behaviour is consistent with the non-unique nature of aerosol optical signatures.

Scattering and SAE respond to particle number, particle size, hygroscopicity, refractive index, and mixing state, so physically different aerosol perturbations can produce overlapping optical responses.[16,17,19] Thus, the optical variables answered whether an aerosol perturbation occurred, but not reliably which event type occurred.

PM-mass responses strengthen with particle size

Size-resolved PM mass showed a stronger size dependence than the optical variables (Figure 2). Event detection increased from PM_1 mass to $PM_{1\text{–}10}$ mass and super-PM_{10} mass, defined operationally here as the derived mass fraction above 10 μm. PM_1 mass had $\Delta^*_{\text{det}} = 1.01$ and a median-difference confidence interval of 1.55–3.88 μg m^{-3}. $PM_{1\text{–}10}$ mass had $\Delta^*_{\text{det}} = 1.64$ and a confidence interval of 1.28–7.07 μg m^{-3}. The strongest event-detection response occurred for super-PM_{10} mass, with $\Delta^*_{\text{det}} = 3.72$ and a confidence interval of 1.56–5.68 μg m^{-3} (Table S5). These PM-mass statistics were calculated from the merged optical–PM-mass dataset, in which each of the main PM-mass variables had 1563 valid observations (Table S4).

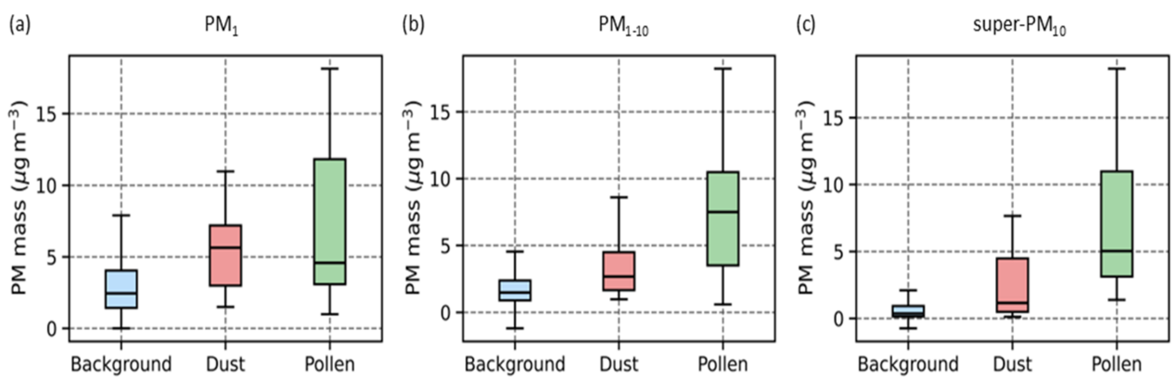

Figure 2: Size-resolved PM-mass response during background, dust-event, and pollen-event conditions in the merged optical–PM-mass analysis. Panels show (a) PM_1 mass, (b) $PM_{1\text{–}10}$ mass, and (c) super-PM_{10} mass, defined operationally as the derived mass fraction above 10 μm. Boxes show the interquartile range, horizontal lines indicate medians, and whiskers show the 5th–95th percentile range. Event-related PM-mass enhancement strengthens with particle size, and the $PM_{1\text{–}10}$ mass contrast provides the clearest mass-based dust–pollen typing signal in the present event set.

Dust–pollen typing showed a different pattern from event detection. PM_1 mass and PM_{10} mass did not robustly distinguish dust from pollen because their dust–pollen confidence intervals crossed zero (Table S6). The clearest mass-based typing signal occurred for PM_{1-10} mass, where the dust–pollen contrast was negative: $\Delta^*_{\mathrm{typ}} = -3.25$, with a median difference of -4.83 μg m^{-3} and a confidence interval of -7.26 to -0.53 μg m^{-3} (Table S6). Because the contrast is defined as dust minus pollen, this indicates higher PM_{1-10} mass during pollen events than during dust events in the present event set. Super-PM_{10} mass showed a similar negative tendency, but its bootstrap confidence interval slightly crossed zero, so it is interpreted as a distributional tendency rather than as a clean median-shift result.

This size dependence is physically consistent with the approximate cubic scaling of particle volume, and therefore mass for particles of comparable density and morphology, with particle diameter.[17] Coarse and supermicron particle changes can therefore produce large PM responses even when optical variables remain only weakly discriminatory. Negative derived mass values occurred only under background conditions and did not drive the PM-mass conclusions (Sect. S6 and Table S7). Excluding negative derived mass values produced only minor changes in the effect-size estimates, including a change in super-PM_{10} detection Δ^* from 3.72 to 3.58 and a change in PM_{1-10} typing Δ^* from -3.25 to -3.24 (Table S8).

Accumulation-mode cross-sectional area shows the clearest APSD-based contrast

The weak source specificity of the optical variables and the stronger size dependence of the PM-mass response indicate that event-type information is likely contained in the redistribution of particles across size, rather than in bulk optical enhancement alone. We therefore examined APSD-derived number, cross-sectional-area, and volume metrics. Both event types modified the aerosol population, but the most persistent dust–pollen contrast occurred in accumulation-mode metrics rather than in bulk optical intensity or coarse-mode volume alone.

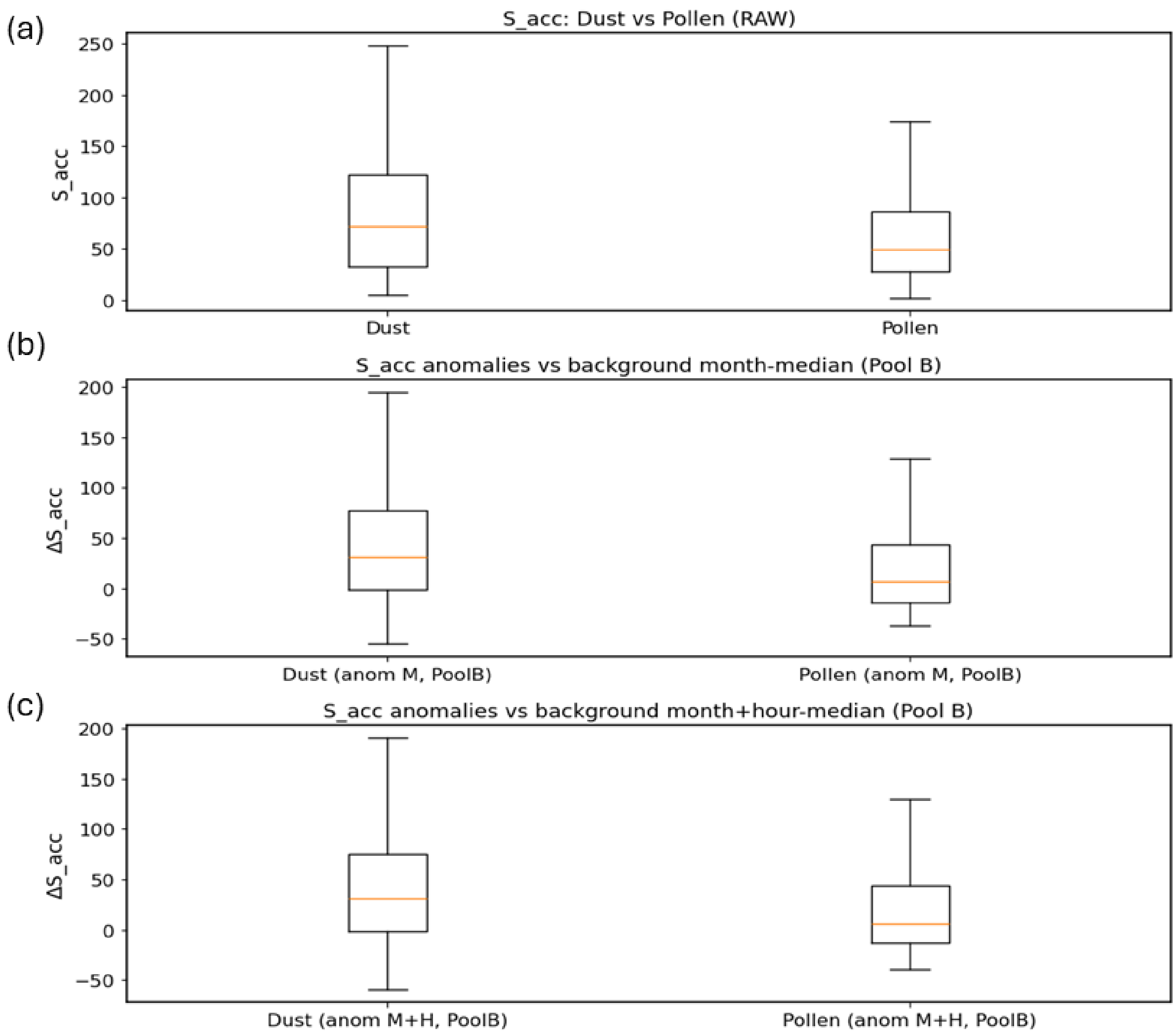

Figure 3: Confounding-controlled dust–pollen comparison for accumulation-mode cross-sectional area, S_{acc}. Panel (a) shows the raw dust–pollen comparison. Panel (b) shows anomalies after subtracting the background month median, and panel (c) shows anomalies after subtracting the background month-plus-hour median. The persistence of the native-resolution dust–pollen difference after month-plus-hour correction suggests that the S_{acc} contrast is not explained solely by seasonal or diurnal background variability. Event-declustered uncertainty is reported in Table S11 and motivates cautious interpretation.

Under the Pool B month-plus-hour anomaly baseline (Figure 3), dust events had higher S_{acc} anomalies than pollen events, with $\Delta^* = 0.558$ and a median anomaly difference of 25.39 $\mu\mathrm{m}^2\ \mathrm{cm}^{-3}$. The corresponding confidence interval, 17.10–31.98 $\mu\mathrm{m}^2\ \mathrm{cm}^{-3}$, did not cross zero, indicating that the controlled S_{acc} contrast persisted after accounting for month and hour background structure at native time resolution (Table S10). After event declustering, the contrast retained the same positive direction and had a larger standardized effect, but the bootstrap confidence interval crossed zero (Table S11). Therefore, S_{acc} is interpreted as the clearest APSD-based contrast in this dataset, rather than as a fully validated source classifier.

Accumulation-mode volume, V_{acc}, showed a consistent supporting response, with $\Delta^* = 0.520$, a median anomaly difference of 1.00 $\mu\mathrm{m}^3\ \mathrm{cm}^{-3}$, and a confidence interval of 0.686–1.238 $\mu\mathrm{m}^3\ \mathrm{cm}^{-3}$ in the native-resolution Pool B month-plus-hour anomaly analysis (Table S10). However, after event declustering, the V_{acc} dust–pollen confidence interval crossed zero (Table S11). Thus, V_{acc} supports the accumulation-mode interpretation but is treated as secondary evidence.

By contrast, coarse-mode volume, V_{coarse}, did not provide a robust dust–pollen separation. Its Pool B month-plus-hour anomaly confidence interval crossed zero, and the standardized effect was small at native time resolution (Table S10). The event-declustered analysis and moving-block-bootstrap sensitivity check also gave confidence intervals crossing zero (Table S11). The volume-ratio metric, R_V, was statistically separated in the native-resolution analysis but had a small standardized effect, and the event-declustered and moving-block-bootstrap sensitivity checks crossed zero (Tables S10 and S11). It is therefore best interpreted as secondary supporting information rather than as a primary discriminator.

The apparently different directions of the $\mathrm{PM}_{1\text{–}10}$ mass and S_{acc} results are not contradictory because the two variables weight the size distribution differently. PM mass is approximately volume-weighted and scales with particle diameter cubed, so $\mathrm{PM}_{1\text{–}10}$ mass can be strongly influenced by larger supermicron and coarse-tail particles. In contrast, S_{acc} is an accumulation-mode cross-sectional-area metric that scales with particle diameter squared and is more directly linked to scattering-relevant particle geometry. Thus, pollen events can show higher $\mathrm{PM}_{1\text{–}10}$ mass in the present event set, while dust events show higher controlled accumulation-mode cross-sectional area. This reconciliation is summarized in Sect. S8. The strongest source-type information was therefore not in bulk optical intensity alone, but in how aerosol area and mass were redistributed across particle size.

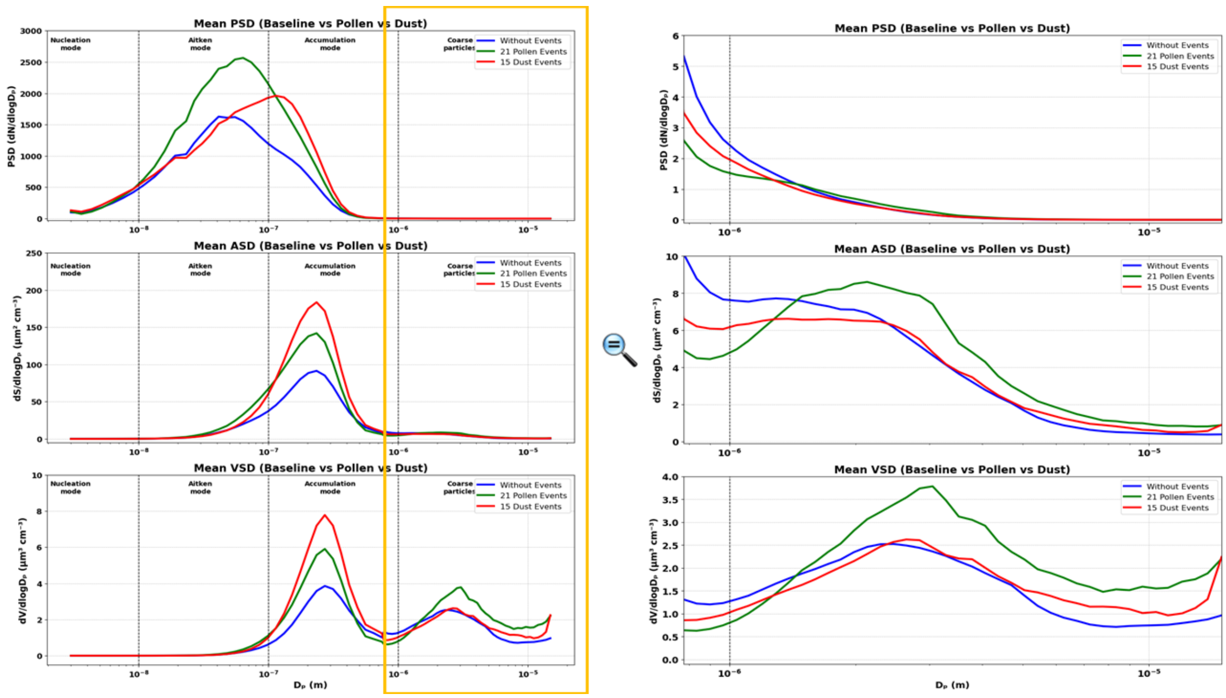

Figure 4: Mean particle size distributions during background, pollen-event, and dust-event conditions. The left panels show the full measured size range for the particle number size distribution, cross-sectional-area size distribution, and volume size distribution. The highlighted region indicates the coarse and supermicron size range, which is shown in the corresponding zoomed panels on the right. In the figure legend, “Without Events” denotes the background class, defined as non-event observations classified as neither dust nor pollen. The comparison illustrates that dust and pollen events can produce overlapping bulk optical responses while redistributing particle number, cross-sectional area, and volume differently across particle diameter. The area-weighted response is most directly linked to S_{acc}, whereas the volume-weighted response helps explain the stronger sensitivity of PM mass to larger particles. Native-resolution APSD medians and IQRs are reported in Table S9.

The mean size distributions further illustrate why the optical and PM-mass responses are not equivalent (Figure 4). In the full size range, pollen periods showed enhanced particle number in the Aitken-size range, whereas dust periods showed a stronger accumulation-mode cross-sectional-area response. In the coarse and supermicron zoom, differences in the volume distribution became more visible, consistent with the stronger sensitivity of PM mass to larger particles. This supports the interpretation that $PM_{1\text{–}10}$ mass and S_{acc} need not show the same dust–pollen direction because they weight the particle size distribution differently.

Sub-100 nm behaviour provides process context rather than source typing

Although the clearest dust–pollen separation was found above 100 nm, the sub-100 nm size range was examined to determine whether event periods also differed in NPF-related particle behaviour, which is common at SMEAR II and has been analysed extensively in long-term event and nonevent classifications.[4,6,7] The 100 nm diameter was used as a diagnostic division, not as a strict physical threshold. It approximately separates the nucleation- and Aitken-mode size range, where NPF, early growth, and survival are most relevant, from the larger accumulation- and coarse-mode size range, where particles contribute more strongly to aerosol cross-sectional area, PM mass, scattering, and potential cloud condensation nuclei (CCN) relevance. NPF survival depends on the competition between particle growth and losses to the pre-existing aerosol population, while CCN activation depends not only on particle size but also on hygroscopicity, composition, mixing state, and supersaturation.[7,8,11,12,15,17,18]

Pollen events showed higher raw Aitken-mode number concentration, N_{Ait}, than dust events, whereas dust-event N_{Ait} remained closer to background (Table S9). However, this contrast weakened after month and month-plus-hour background correction, indicating that part of the raw difference was linked to the seasonal and diurnal conditions under which pollen events occurred (Tables S10–S12). NPF-like behaviour occurred during both dust and pollen events, and the extended sub-100 nm diagnostics are provided separately for pollen and dust events in Figs. S1–S6. Therefore, sub-100 nm particle behaviour should be interpreted as process context rather than as a stand-alone dust–pollen discriminator.

Implications

These results have three implications for interpreting boreal aerosol event records. First, bulk optical enhancement is not sufficient evidence for event type. Dust and pollen both enhanced aerosol scattering, and their optical signatures overlapped. Source typing based only on $\sigma_{\mathrm{sca},550}$ or SAE could therefore be misleading.

Second, size-resolved information is essential. Event occurrence was most strongly detected in larger PM fractions, especially super-PM_{10} mass, while event type was better expressed in $\mathrm{PM}_{1\text{–}10}$ mass and accumulation-mode APSD metrics. Among the APSD metrics, the clearest and most persistent dust–pollen contrast was found in S_{acc}, supported by V_{acc}, rather than in coarse-mode volume. This indicates that transported dust and pollen influence the receptor-site aerosol population through different redistributions of accumulation-mode area and volume, even when their bulk optical responses overlap.

Third, NPF-related sub-100 nm behaviour should be interpreted as process context rather than as a primary event-typing signal. Event-day sub-100 nm structure can reveal whether nucleation- and Aitken-mode processes were active, but the strongest dust–pollen information in this analysis was found above 100 nm.

Conclusions

This study was structured around three questions: how dust and pollen events affect the boreal aerosol population, which variables detect event occurrence relative to the non-event background, and which variables distinguish dust events from pollen events.

For the first question, dust and pollen events both perturbed the aerosol population at SMEAR II, but their effects were not expressed in the same way across all aerosol properties. Both event types enhanced aerosol scattering relative to background conditions, showing that they modify the in-situ optical aerosol population. However, the particle-size-distribution and PM-mass responses showed that the event influence was not limited to bulk optical enhancement. Instead, dust and pollen events redistributed aerosol number, cross-sectional area, volume, and mass differently across particle size.

For the second question, event detection was clearest in variables that captured enhanced particle loading, especially in the larger PM fractions. Optical scattering coefficients detected event-related perturbation, but the strongest detection response occurred for super-PM_{10} mass. This shows that the occurrence of dust or pollen events at the receptor site is most clearly expressed through size-resolved mass enhancement, particularly in the coarse and supermicron size range.

For the third question, the variables that detected events were not necessarily the variables that distinguished dust from pollen. Scattering coefficients and scattering Ångström exponents showed substantial dust–pollen overlap and therefore did not provide robust source typing. Among the PM variables, $PM_{1\text{–}10}$ mass provided the clearest mass-based dust–pollen contrast in the present event set. Among the APSD-derived variables, accumulation-mode cross-sectional area, S_{acc}, showed the clearest temporally controlled dust–pollen contrast, with support from accumulation-mode volume, V_{acc}. Coarse-mode volume, V_{coarse}, did not robustly distinguish event type. These results indicate that dust–pollen typing was expressed more clearly in the size-resolved redistribution of aerosol area and mass than in bulk optical intensity alone.

The analysis also showed that sub-100 nm particle behaviour provides process context rather than a primary event-typing criterion. NPF-like behaviour occurred during both dust and pollen events, and raw Aitken-mode differences weakened after temporal background correction. Therefore, sub-100 nm variability helps describe the aerosol processes occurring during event periods, but it does not by itself provide a robust basis for separating dust from pollen.

Overall, the answer to the three guiding questions is that dust and pollen both perturb the boreal aerosol population, event occurrence is best detected through enhanced size-resolved mass and scattering, and event type is better resolved through particle-size-distribution metrics than through bulk optical enhancement. Distinguishing dust from pollen in boreal in-situ aerosol observations therefore requires size-resolved aerosol information and cannot rely on optical enhancement alone. Because of the small number of independent event episodes, especially for pollen, the results should be interpreted as physically interpretable event-type tendencies rather than as a definitive operational classification model.

Supporting Information Available

The Supporting Information PDF contains the detailed event-classification and event-count logic; event-month distribution; optical-only and merged optical–PM-mass observation counts; variable-specific event counts; measurement site, instruments, and derived-variable descriptions; statistical definitions; full merged optical–PM-mass event-detection and dust–pollen typing effect-size tables; negative derived mass-value audit and sensitivity tests; APSD-derived metric counts, medians, native-resolution anomaly contrasts, event-declustered and moving-block-bootstrap sensitivity checks; reconciliation of PM_{1-10} mass and S_{acc}; and extended sub-100 nm and NPF-related diagnostics, including particle number, volume, and cross-sectional-area size-distribution figures for pollen and dust events in Figs. S1–S6.

Author Contributions

S.B., V.-M.K., and T.P. conceptualized the study. S.B. and T.G. performed the data analysis. S.B. wrote the original draft and prepared the manuscript with input from V.-M.K. V.-M.K. and T.P. contributed to the review and revision of the manuscript, including the final edits. T.P. provided overall supervision and secured funding.

Data Availability

SMEAR II observational data are subject to the station data policy and data-access procedures. Processed analysis tables and scripts supporting the results will be made available through a public repository or from the corresponding author according to the final data-access arrangement at submission.

Notes

The authors declare no competing financial interest.

Acknowledgments

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